

Chains of reasoning - Set 2

TMUA practice questions by topic. Specification Prf2, Prf3, Prf5. 11 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?
- A** 4
B 3
C 0
D 1
E 2
F infinitely many
- 2 The product of three consecutive positive integers is N , and the middle one of the three integers is m . Which of the following statements must be true?
- I** N is divisible by 6.
II N is divisible by 4.
III $N + m$ is a perfect cube.
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III

- 3 How many positive integers less than 1000 are divisible by neither 2 nor 5?
- A 399
 - B 600
 - C 500
 - D 400
 - E 450
 - F 300
- 4 p and q are prime numbers with $p + q = 45$. Find pq .
- A 247
 - B 140
 - C 102
 - D 86
 - E 94
 - F 50
- 5 A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?
- I The sum of all 10 terms is negative.
 - II The sequence contains at least one negative term.
 - III The sequence contains at least one positive term.
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III

- 6** Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?
- A** Ben
 - B** Alice
 - C** Cara
 - D** It cannot be determined.
 - E** The situation is impossible.
- 7** A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?
- A** 16
 - B** 20
 - C** 14
 - D** 8
 - E** 17
 - F** 12
- 8** n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?
- A** $n = 3$
 - B** n is prime
 - C** n is odd
 - D** n is a perfect square
 - E** n is even

- 9** The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?
- I** $a \leq 3$
II $c \geq 5$
III $b = 4$
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III
- 10** The integers a , b , c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?
- I** Exactly one of a , b , c is positive.
II $a^2 + b^2 + c^2$ is even.
III One of a , b , c is divisible by 3.
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III

11 How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

A 4

B 3

C 2

D 6

E 8

F 5